	assumption in **(b)(i)** is valid.
5(a)	1. There is no net flow of energy in a stationary wave. 2. The wave particles in a stationary wave are oscillating either in phase or in antiphase with one another at varying amplitudes.
5(b)	When the wave from the source approach the plate, it is reflected by the plate. Since the wave from the source that move normally to the plate and the reflected wave have the same amplitude, frequency and are moving opposite in direction, they interfere with each other to form stationary waves.
5(c)(i)	The waves from the point source radiate radially outwards, which results in the intensity and amplitude of the waves decreasing with distance propagated, even after reflection at the plate. Since the amplitude of the waves from the source and the reflected waves are not the same, there is no complete destructive interference. This means that the net amplitude, and hence the intensity, of the waves will not be zero.

5(c)(ii)	$\lambda = 2(3.6 - 1.0) = 5.2$ cm
5(c)(iii)	$v = f\lambda$ $f = \dfrac{3.0 \times 10^{8}}{5.2 \times 10^{-2}} = 5.77 \times 10^{9}$ Hz
6(a)	It is the amount of time required for a quantity of the radioactive isotope particles to decay